



XS Sheet Numbers

xs16-128-1, xs16-128-2, xs16-128-3, xs16-128-4, xs16-128-5, and xs16-128-6

Technical drawing of a sidewalk cross-section showing dimensions and components. The drawing includes the following labels and dimensions:

- Overall Width:** 8'-2"
- Top Section:** 3" (width), 1'-5" (width), 1'-3" (width), 9" (height), 2" (height), 3/4" CHAMFER, Typ (corner chamfer).
- Post Section:** 10" (height), 7" (width), 8" (width), TO BALUSTER (text), 1" (width), POST FACE (text).
- Optional Section:** OPTIONAL Const JT (text), 3" (width), 9/2" (height).
- Post Face Section:** 1'-6" (height), 1'-2" (height), 3'-6" (height), 1'-0" Min (height).
- Sidewalk Section:** SIDEWALK (text), 1.5% (slope), 2" (width), 8" (width).
- Bridge Deck Section:
 - BRIDGE DECK (text)
 - 8" Min (height)
 - 2'-0" (width)
 - DRIP GROOVE (text)
 - 8" Min (height)**
- Reinforced Concrete Box Girder Bridge Section:** REINFORCED CONCRETE BOX GIRDER BRIDGE (text).

Figure 1 Typical Section of Concrete Barrier Type 86HSW

The Concrete Barrier Type 86HSW bridge railing (TYPE 86HSW) complies with the National Historic Preservation Act and for use on projects that may require consultation with the State Historic Preservation Officer (SHPO) in the California Department of Parks and Recreation. Concrete Barrier Type 86HSW includes an aesthetic component between upper beam and lower beam (curb) composed of concrete balusters and clear openings between balusters that are visually compatible with the open concrete baluster rail seen on many historic bridges. The “H” in Type 86HSW stands for “Historic” because its most common use will be at locations where the existing structure has a historic designation, or the area is a historic district.

This Concrete Post & Beam see-thru Combination Railing (vehicular and pedestrian) is approved per MASH 2016 (AASHTO Manual for Assessing Safety Hardware) and



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approved for TL-2 low-speed locations (regulatory speed limits of 45 mph or less). This Type 86HSW is a see-thru Concrete Post and Beam style concrete barrier with concrete lower beam (curb) over integral sidewalk that is connected to reinforced concrete box girder bridge deck, wingwall, retaining wall as shown on the plans. It can be specially designed to connect to concrete barrier moment slab, structure approach slab, or top slab of concrete box culvert where the top slab acts as the vehicular riding surface.

Geometric data from bottom of concrete lower beam (curb) and above for parapet portion of the Concrete Barrier Type 86HSW matches the Concrete Barrier Type 86H, the integral raised concrete sidewalk is the new added feature. Vehicular rail height is 3 feet – 6 inches above top of integral raised concrete sidewalk. This Bridge Rail meets the 42-inch minimum pedestrian railing height without adding tubular handrailing. Pedestrian tubular handrailing can be added as shown, if desired. The sidewalk curb is 8-inches above the Finish Grade of either the bridge deck or the deck overlay if an overlay is to be placed on the same contract. From the traffic face of the sidewalk curb, the top of the sidewalk slopes up at 1.5% toward the edge of deck (EOD). The 1.5% cross slope of the top surface of the sidewalk gives 0.5% construction tolerance to ensure that the constructed integral raised sidewalk of the Concrete Barrier Type 86HSW does not exceed the maximum cross slope of 2% per ADA law and accessibility guidelines and policies.

Overall barrier width is 2 feet – 0 inches.

Post spacing is 10 feet – 0 inches maximum. Post spacing between BB and EB should be equal spacing. Post spacing on wing walls may be less than on bridge deck. Minimum post spacing between post centerlines is 5 feet - 0 inches.

Aesthetic see-through railing.

Standard Drawing Features

All six of the Bridge Standard Detail Sheets must be reviewed by the PDT and any historical committees representing the community for applicability, then pertinent details may be edited as needed for a specific project by the Engineer of Record (EOR) and included in the contract plans:



User Guide to Bridge Standard Detail Sheets

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Sheet Number: xs16-128-1

Includes typical railing system layout and concrete barrier reinforcement. Typical section of the concrete railing on concrete lower beam (curb) with approach and departure concrete transition end blocks, and design notes.

Sheet Number: xs16-128-2

Includes railing system details on top of a wall or trench footing, some transition end block details, and a perspective view.

Sheet Number: xs16-128-3

Includes reinforcement details in the elevation view, and plan views of upper beam and lower beam (curb), and at expansion joint locations.

Sheet Number: xs16-128-4

Includes basic baluster alignment & reinforcement details, plus architectural options for baluster shapes and for shapes at top of clear openings between balusters.

Sheet Number: xs16-128-5

Includes details for tubular pedestrian railing, HSS steel tube standard splice, and details for HSS steel tube expansion splice.

Sheet Number: xs6-128-6

Includes MASH compliant details for approach end block details and features the vertical slotted holes to aid with constructability of the three beam rail. Caltrans is adopting bridge approach end block per the crash tested and approved details developed by the Midwest Roadside Safety Facility (MwRSF) at the University of Nebraska (TRP 03-367-19-R1) for the MwRSF Pooled Fund of which Caltrans is a member State DOT.

The end of the approach end block tapers down in height to 35" above top of the integral concrete sidewalk.

Design/General Notes

Design Criteria:



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AASHTO LRFD Bridge Design Specifications 8th edition with California Amendments (AASHTO-CA BDS-8):

Live Loading

- HL 93 and permit design load Vehicular Collision Force
- MASH Test Level 4

Concrete

- $f_y = 60$ ksi (ASTM A706/706M, Grade 60)
- $f'_c = 3.6$ ksi

Steel Components for pedestrian tubular handrailing

- Carbon steel structural rail, and post tubing (HSS): ASTM A500/A500M, Grade B
- Carbon steel plate and splice sleeves: ASTM A36/A36M
- Bolts: ASTM F3125, Grade A325/A325M, Type 1
- Threaded rods: ASTM A449, Type 1 or ASTM F1554, Grade 55, Class 2A
- Nuts for bolts and threaded rods: ASTM A563/A563M
- Washers for bolts and threaded rods: ASTM F436/F436M

Designers must ensure that any supporting structures, such as the bridge deck, wing wall, retaining wall or bridge deck overhang, meet the requirements in the AASHTO LRFD Bridge Design Specifications, Appendix in Section 13, Railing, as amended by Caltrans California Amendments (AASHTO-CA BDS-8 – Section 13).

Supporting elements, such as the deck and overhang, must be designed to three applicable load cases:

- Case 1: Extreme Event II (transverse and longitudinal forces)
- Case 2: Extreme Event II (vertical forces)
- Case 3: Strength I

The clearance to reinforcement in the concrete lower beam (curb) is 2 inches at the traffic face and back face, and 1 inch on the top. Clearance to reinforcement in the integral raised concrete sidewalk is 2 inches. The clearance to reinforcement in the concrete transition end blocks at the approach and departure ends is 1 inch at the traffic face, the back face, and the top.



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For projects located in a corrosive environment, refer to AASHTO-CA BDS-8, Section (5.10) for using epoxy coated rebar and Standard Specifications section 52-2.

Concrete Barrier Type 86HSW was designed and checked as a new MASH-compliant bridge railing and its design is based on Type 86H.

Sidewalk Design:

Per the Caltrans Highway Design Manual (HDM), Section 208.4 Bridge Sidewalks:

“Sidewalks on bridges should be provided wherever there are sidewalks or other pedestrian facilities that follow the highway. The minimum width of a bridge sidewalk shall be 6 feet. The recommended width should be 8 feet for pedestrian comfort. Bridge sidewalks in area types (see Index 81.2) with high levels of pedestrian activity may need to be greater than 8 feet.”

The 6 feet minimum width of bridge sidewalk noted in Section 208.4 of the Highway Design Manual refers to the walking surface at the top of the sidewalk and does not include the 2-inch width for the slope of the sidewalk curb face next to the roadway shoulder. Similarly, the recommended 8 feet sidewalk width noted in Section 208.4 of the Highway Design Manual refers to the walking surface at the top of the sidewalk and does not include the 2-inch width for the slope of the sidewalk curb face next to the roadway shoulder.

Crashworthiness:

Concrete Barrier Type 86HSW was designed and checked as a new MASH-compliant bridge railing; its crashworthiness is based on the Concrete Barrier Type 86H Crash Test.

A link to the site for Vehicular Crash Tests of the Concrete Barrier Type 86H Research Results will be updated after the crash test report is posted online. The Final Crash Test Report and other general information will be posted to the Division of Research and Innovation and Systems Information webpage for [Research Final Reports in the Geotechnical and Structures subsection](#). An FHWA Letter of Eligibility is no longer required by FHWA, but one will be requested after approval and posting as a matter of formality and to get this concrete barrier added on the FHWA list of approved MASH concrete barriers. More information regarding MASH Implementation by Caltrans can be found at the Caltrans/Division of Safety Programs [Implementation of the Manual for Assessing Safety Hardware \(MASH\)](#) webpage.



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Pedestrian Accessibility:

Concrete Barrier Type 86HSW Bridge Rail complies with the pedestrian accessibility guidelines in [Design Information Bulletin 82 \(DIB 82\)](#).

Bridge deck joints shall continue through the barrier and must be armored to comply with the requirements in DIB 82 for allowable change in horizontal and vertical elevations in pedestrian walkways. See, “Joint Armor for Pedestrian Walkways” detail on [Bridge Standard Details, xs8-050 sheet](#) and the [Bridge Standard Details, xs8-050 User Guide](#).

If the Maximum Movement Rating is more than 2 inches, contact the Caltrans Bridge Joints and Bearings Specialist or email at DESdesign@dot.ca.gov with “Attention: Caltrans Bridge Joints and Bearings Specialist”.

Utilities and Overlays:

Only two 1 ½ inches diameter conduits are permitted in the taller half of the concrete curb of the concrete barrier parapet on the raised integral sidewalk (front-to-back curb height of parapet from 18 inches to 15 inches).

Conduits are permitted in the integral raised sidewalk of the bridge rail as noted on the Bridge Standard Detail Sheets. As a minimum, provide two 4-inch round openings, ducts, or conduits for future use as shown on the plans. In regard to conduits at the end of bridge or wingwall see Standard Plans B14-3, ES-9A, and ES-9B.

Designers shall consider cross-slope of the sidewalk, super elevation of the bridge deck, whether an overlay is going to be placed in the same contract on the bridge deck in front of Concrete Barrier Type 86HSW integral raised sidewalk (in which case the sidewalk must be constructed deeper so that the height of the sidewalk curb will be a minimum 8 inches above the top of the Finish Grade of the deck overlay). Designers shall also consider the minimum distance that conduits must be clear of the sidewalk curb face per the Bridge Standard Detail Sheets for Type 86HSW, the minimum distance that conduits must be clear of the traffic face of the vertical parapet portion of the concrete barrier per the Bridge Standard Detail Sheets for the Type 86HSW, the minimum 2 inches of clear space required between the outside edges of adjacent conduits, and other factors to ensure that there is adequate cover over and adequate spacing of planned or future conduits. Due to these considerations, designers may have to reduce the size and or number of conduits from the maximum allowable number and size of conduits shown in the NOTES on the Bridge Standard Detail Sheets for the



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Type 86HSW. For each additional foot of sidewalk width above the minimum sidewalk width noted on the Bridge Standard Detail Sheets, one additional 4-inch nominal diameter (or smaller) conduit can be added.

If a deck overlay is being added to the bridge deck or approach slab on the same contract that the concrete barrier is being constructed, then the concrete sidewalk curb should be constructed to an additional height equal to the depth of the overlay (not to exceed 2 inches in additional height). After the overlay is placed, the height of the concrete curb is 8 inches above the Finish Grade (FG) of the deck overlay instead of the concrete bridge deck, and the height of the vehicular railing parapet on the integral raised sidewalk will measure 3 feet – 6 inches above the top of the sidewalk as shown on the plans.

If a deck overlay is planned for an existing bridge deck with an existing Concrete Barrier Type 86HSW, then consider the following options:

- No deck overlay.
- Taper the deck overlay down to the minimum depth permissible and stop at least 3 feet – 0 inches away measured transversely from the traffic side toe of the integral raised concrete sidewalk curb of the Concrete Barrier Type 86HSW.
- If need deck overlay to extend all the way to the existing curb face such as in a marine environment or in snow country where it is needed to seal the deck surface, then taper down the depth of the overlay starting at the Edge of Travelled Way down to the a depth of 2" or less so that the height of the sidewalk curb at the traffic face is no less than 6" height above the top of the overlay Finish Grade. If this is not possible, then may have to replace the existing Type 86HSW in conjunction with an overlay placed all the way to the toe of the sidewalk curb in which case the overlay depth would be added to the height of the raised integral concrete sidewalk of Type 86HSW so that the integral sidewalk curb height above the top of the overlay Finish Grade is 8 inches (contact SM&I Load Rating Unit to determine if additional weight will change load rating factor or permit load level of existing bridge).
- If the shoulder is narrow which leaves little or no distance to taper down the deck overlay depth, then choose an overlay material that can be applied in a depth of 2 inches or less in order to preserve a minimum of 6 inches sidewalk curb height of the existing Concrete Barrier Type 86HSW.



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Additional Drawings Needed to Complete PS&E

If the Type 86HSW concrete transition end blocks for a project are going to connect to something other than the guardrail transition Standard Plans for either Thrie Beam Barrier guardrail or Midwest Guardrail System, then special designed detail drawings will be required.

If the tubular handrailing is needed to be taller than 48 inches in height, then special designed detail drawings will be required. Additional detail drawings will be required if there is going to be any architectural texture on the back side and/or traffic side of the bridge rail.

Contract Specifications

Caltrans Standard Specifications: Section 51 Concrete Structures, Section 52 Reinforcement, Section 55 Steel Structures, Section 59 Structural Steel Coatings, Section 75 Miscellaneous Metal, Section 83 Railing and Barriers, Section 91 Paint, and if a special design is done to add chain link railing to Type 86HSW (such as, if the bridge goes over railroad tracks or if bridge is in an urban area or where a large volume of pedestrian traffic is anticipated, or other factors that may determine the need for chain link railing) then also Section 83-1.021 Chain Link Railing..

Restrictions on Use of Standard Drawings

- A special design is required if Concrete Barrier Type 86HSW is mounted directly to the top of an Earth Retaining System (ERS) such as soldier pile wall, tie-back wall, or soil nail wall. For MSE walls, which are one type of ERS, a Type 86HSW would have to be mounted on concrete barrier moment slab over the top of the MSE wall (see xs12-090, xs13-020-5, and xs13-020-6).
- Sound walls cannot be mounted on the Concrete Barrier Type 86HSW.
- A special design is required to mount a chain link railing to the Concrete Barrier Type 86HSW. Chain link railing should only be added to Type 86HSW for specific lengths where required over railroad tracks or where over or adjacent to locations with security concerns.
- A special design is required for retrofitting this Type 86HSW onto an existing bridge deck, existing retaining wall, existing approach slab, top slab of existing concrete box culvert or existing barrier moment slab. Due to the need for long hooks as well as the additional transverse deck bars at each post location and due to the existing bridge deck design and deck overhang size, Type 86HSW



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may not work as a retrofitted concrete barrier onto an existing bridge without removal and replacement of the existing bridge deck overhang. If the Concrete Barrier Type 86HSW is desired to be added to an existing bridge and if the additional transverse bridge deck bars required at the post locations cannot reach the required minimum 2 feet – 0 inches inward from the exterior girder with a deck overhang removal and replacement, then carbon-fiber-reinforced-polymers (CFRP) near surface mounted deck strengthening needs to be considered. If finite element analysis indicates that the Concrete Barrier Type 86HSW can be retrofitted onto the existing deck/deck overhang, the analysis may show that CFRP near surface mounted deck strengthening will be required before placement of the concrete barrier. The Concrete Barrier Type 86HSW cannot be retrofitted onto the top of an existing retaining wall unless the existing retaining wall was designed for the transfer of vehicular impact loading, or a full design check is done that shows the existing wall design will be adequate for the transfer of vehicular impact loading. The top of the existing wall will need to be removed and replaced with a haunch with adequate reinforcement when the existing wall design can handle the impact load. If the existing retaining wall is incapable of handling the vehicular impact loading, then either the Concrete Barrier Type 86HSW will have to be mounted on a concrete barrier moment slab or be mounted on a structure approach slab that extends over the top of the existing retaining wall or wing wall.

- This concrete barrier with integral raised sidewalk was designed with the pedestrian railing attached (offset 9 inches from the top of the traffic-side of the vertical vehicular parapet portion of this concrete barrier with integral raised sidewalk).
- The clear openings for the vehicular rail and for the pedestrian railing are in conformance with the size limits set forth in Section 13 of the AASHTO LRFD Bridge Design Specifications along with Section 13 of the California Amendments

Special Considerations

Vehicular Barrier:

If one side of the bridge has Type 86H (42-inch vehicular rail height) and the other side has low-speed traffic then Type 86HSW should be used for aesthetics (42-inch vehicular railing height on integral concrete sidewalk). If the existing bridge has a 42-



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inch or greater height see-through aesthetic concrete baluster bridge rail mounted on integral concrete sidewalk (that does not meet current standards and/or is damaged or deteriorated), and the new project is either replacing the bridge rail or replacing the bridge, and -to address any context sensitive issues- it is desired to replace the bridge rail with a see-through aesthetic concrete baluster bridge rail that meets current standards ... then the see-through aesthetic 42-inch vehicular rail height Concrete Barrier Type 86HSW would be a good option.

If the specific structure location with associated roadway alignment and profile grade does not require the pedestrian tubular handrailing height to be higher than the required minimum of 42-inches, then the pedestrian tubular handrailing shown in the Bridge Standard Detail sheets mounted on top of the Type 86HSW can be omitted because the vehicular bridge railing height of 42 inches already meets the minimum tubular handrailing height requirement.

Combination of Pedestrian Tubular Handrailing/Bicycle Railing:

The Highway Design Manual (HDM), Section 208.10(5) and Section 1003.1(3), states that a sidewalk on a bridge can only include bicycle traffic in addition to pedestrian traffic if both the walking surface of the sidewalk is a minimum 10 feet – 0 inches wide and signage is posted that specifically permits bicyclists to share the sidewalk with pedestrians.

If the walking surface of the integral bridge rail sidewalk is 10 feet – 0 inches wide or more and signage is posted that specifically permits bicyclists to share the sidewalk with pedestrians, then typically, the designer should strongly consider using a taller-than-minimum height bicycle railing when there is some combination of these below-listed factors. If the bicycle railing is modified from the standard details shown on the Type 86HSW Bridge Standard Details sheets, then a special design with special modified details will be required. But if any one of them is closer to the extreme end for that particular factor, then that factor alone could determine the need for a taller-than-minimum height bicycle railing. The design considerations where a 48-inch or greater height bicycle railing would be strongly recommended are:

- Bridge located on small radius curve in roadway alignment (in relation to the posted speed limit)
- High super elevation (of 6% or more)
- A roadway shoulder that is less than the minimum required width



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- If the roadway profile grade is a sustained 3% or more (where sustained is 300 feet or more along the traffic lane)
- If the roadway has frequent conditions of low or impaired visibility (fog, sandstorm, limited sight distance, etc.)
- Located at bottom of T-shaped intersection (for this situation a 54-inch bicycle railing is strongly recommended)
- If local stakeholders and the Caltrans District or local agency have agreed on a taller than minimum bicycle railing height for any reason

Aesthetics:

Aesthetic see-through concrete barriers such as the Concrete Barrier Type 86HSW are preferred by the California Coastal Commission for use within the Coastal Zone and may also be selected for any location where a Context Sensitive Solution is warranted.

It's common to customize post-and-beam bridge rails with aesthetic and context sensitive details. Contact the Bridge Railing Technical Specialist in the Caltrans, Division of Engineering Services, Office of Design and Technical Services if a special design is desired.

Regarding aesthetics:

- Balusters can be in multiple shapes (two shapes with either square or chamfered cross-section are shown on the Bridge Standard Detail sheets). Baluster shapes could also be context sensitive special designed and detailed to match or complement some feature on or near the bridge. The designer must choose the baluster shape and show/ note it on the structure plans and delete the remaining baluster shape options and either Section G-G or Section H-H on xs16-128-4 must be deleted as well.
- Clear opening sizes between concrete balusters and between bicycle railing members must comply with AASHTO-CA BDS-8 Section 13.9 Bicycle Railings and Section 13.8 Pedestrian Railing. The Bridge Standard Detail sheet xs16-128-4 shows three standard shapes for the top of the clear openings (gable, squared, and arch). The designer must choose the shape to use for the top of the clear openings and show/ note it on the structure plans and delete the rest of the clear opening shape options. Delete " NOTE: Only one shape option shall be used per structure."
- If a Context Sensitive Solution is desired for the pedestrian tubular handrailing, then a special design could be done for the pedestrian tubular handrailing itself. If



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so, the special design railing must comply with the design capacity and clear opening requirements shown in Section 13 RAILINGS of AASHTO-CA BDS-8.

- Concrete barrier can have color added by either staining the concrete surface or adding dye to the concrete mix (adding dye to concrete mix is the most resistance to blast cleaning and graffiti removal), or both. When adding color to concrete barrier surface, stain should be used. Stain penetrates the surface so if the concrete surface is lightly impacted the color will still remain, whereas paint is only adhered to the surface and will scrape off even if lightly impacted. Paint peels over time and more rapidly in harsh environments.
- Architectural texture can be added to the surface of concrete barriers, but the depth of texture must be added to the outside of the cross section of the standard details for the concrete barrier (so a textured barrier will be wider than the standard barrier and this may affect the bridge width). When texture is added to a concrete post-and-beam bridge rail, the minimum offsets from the upper beam to the posts and from the lower beam (curb) to the posts must both be preserved on the traffic side. If texture is planned for the traffic side of a concrete post-and-beam bridge rail the lower beam (curb) portion of the rail can only have very little texture depth, and if it has any texture it needs to also be a smooth texture design so that tires will not climb the face of the rail. If architectural texture is desired, contact the Bridge Railing Technical Specialist.
- All pedestrian tubular handrailing components require a galvanized coating, galvanized railing can be painted after galvanization. There are no restrictions on choice of coating color for the steel elements, except that yellow cannot be used because the Manual on Uniform Traffic Control Devices (MUTCD) reserves that color for the median striping (cannot have a yellow-colored bicycle railing at outside edge of structure/roadway). Common choices are the galvanized dull grey (unpainted but coated with copper sulfate solution), the galvanized chrome grey (unpainted), Natina Stain (rusty brown or mottled rusty brown) over the galvanized steel railing, or white, light blue, green, black, brown or Golden Gate orange paint over the galvanized steel railing.
- The concrete post spacing must be adjusted to have equal post spacing between bridge joints (for example, between BB and EB along the edge of deck or between bents if the bridge is a steel girder bridge or precast girder bridge that results in joints at each bent) which will result in different concrete baluster locations between concrete posts. The goal is to maximize the number of clear openings between concrete posts. The spacing of concrete balusters and clear openings between the concrete balusters must be centered between the



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concrete posts. The widths of the concrete balusters must be at least 6 inches and should not exceed 6 inches in width unless the desire is to match the width of an existing concrete baluster bridge rail. When the concrete post spacing is adjusted to ensure equal concrete post spacing between bridge joints, then the same adjustment will be required for the pedestrian tubular handrailing post spacing. Due to the length of wing walls and the minimum length of the solid concrete end blocks, the spacing of Type 86HSWA concrete posts will oftentimes be less than 10 feet and 0 inches; the minimum post spacing between posts centerlines shall be 5 feet - 0 inches.

- The vehicular rail height and the pedestrian tubular handrailing height above the top of integral raised sidewalk for the parapet portion of Type 86HSW at the completion of construction contract cannot be less than the heights shown on the Bridge Standard Detail sheets for Type 86HSW (42-inch vehicular rail height and 48-inch minimum pedestrian tubular handrailing height if desired), but the pedestrian tubular handrailing can be taller than 48-inches if desired. If pedestrian tubular handrailing is modified from the standard details shown on the Type 86HSW Bridge Standard Detail sheets, then a specially designed details will be required and will be placed in the same location shown on the plans for the 6-inch height pedestrian tubular handrailing.

Fixed objects, such as lighting standards or OH Signs mounted on a bridge, must be placed on a corbel or pedestal on the back side of a post location of the concrete post-and-beam style Type 86HSW. It's very likely that corbels supporting OH Signs mounted on a bridge will have to locate the corbel at a bent location. Corbel and pedestals will require a special designed detail. For special situations, contact both the Signs and Overhead Structures Technical Specialist and the Bridge Railing Technical Specialist in the Caltrans, Division of Engineering Services, Office of Design and Technical Services by email at [DES Design and Technical Services](#).

If scuppers are desired/needed then contact the Bridge Railing Technical Specialist by email at [DES Design and Technical Services](#). Scuppers through sidewalk are not allowed, also note that deck drainage can only be allowed to drop off the edge of deck by means of scuppers or down-drains if the Environmental Document allows it, and if there are no vehicular lanes/railroad tracks/bicycle path/walkway/or boating waterway directly below.

All project-specific modifications to the Concrete Barrier Type 86HSW must be reviewed by the Bridge Railing Technical Specialist in the Caltrans, Division of Engineering Services, Office of Design and Technical Services. Contact the Bridge Railing Technical Specialist by email at [DES Design and Technical Services](#).